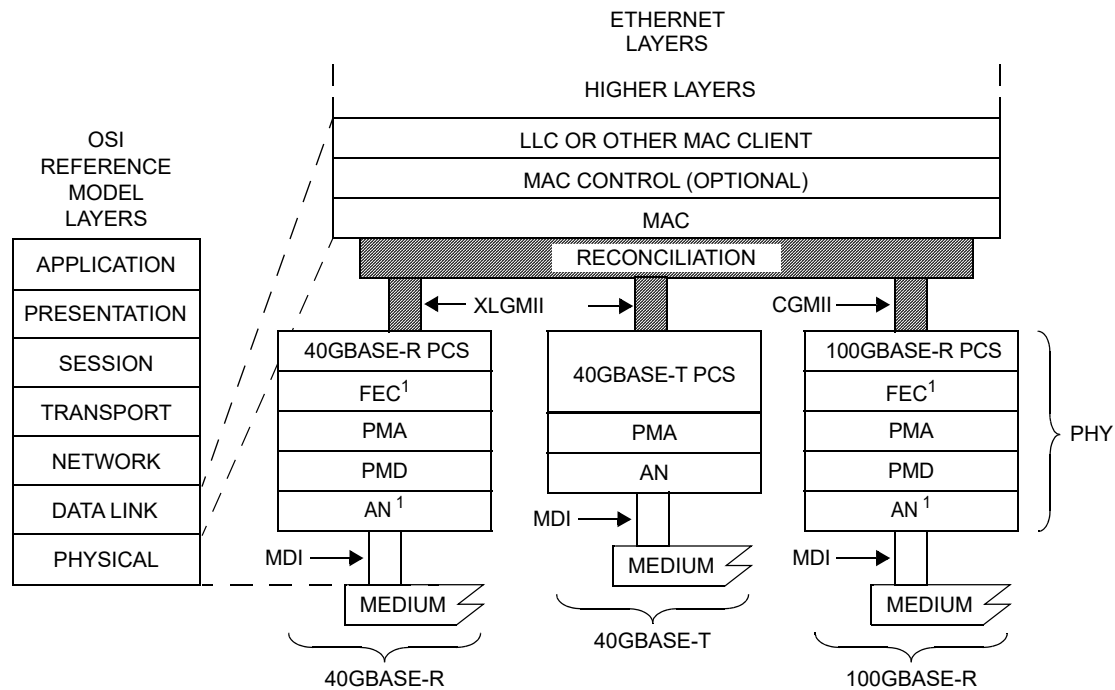


81. Reconciliation Sublayer (RS) and Media Independent Interface for 40 Gb/s and 100 Gb/s operation (XLGMII and CGMII)

81.1 Overview

This clause defines the characteristics of the Reconciliation Sublayer (RS) and the Media Independent Interface between Ethernet media access controllers and various PHYs. Figure 81–1 shows the relationship of the RS and Media Independent Interface to the ISO/IEC OSI reference model. Note that there are two variants of the Media Independent Interface in this clause, the 40 Gb/s Media Independent Interface (XLGMII) and the 100 Gb/s Media Independent Interface (CGMII).



AN = AUTO-NEGOTIATION

CGMII = 100 Gb/s MEDIA INDEPENDENT INTERFACE

FEC = FORWARD ERROR CORRECTION

LLC = LOGICAL LINK CONTROL

MAC = MEDIA ACCESS CONTROL

MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE

PMA = PHYSICAL MEDIUM ATTACHMENT

PMD = PHYSICAL MEDIUM DEPENDENT

XLGMII = 40 Gb/s MEDIA INDEPENDENT INTERFACE

NOTE 1—CONDITIONAL BASED ON PHY TYPE

Figure 81-1—RS and MII relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

The XLGMII and the CGMII are optional logical interfaces between the MAC sublayer and the Physical Layer device (PHY).

The RS adapts the bit serial protocols of the MAC to the parallel format of the PCS service interface. Though the XLGMII/CGMII is an optional interface, it is used in this standard as a basis for specification. The Physical Coding Sublayer (PCS) is specified to the XLGMII/CGMII, so if not implemented, a conforming implementation shall behave functionally as if the RS and XLGMII/CGMII were implemented.

The XLGMII/CGMII has the following characteristics:

- a) The XLGMII supports a speed of 40 Gb/s.
- b) The CGMII supports a speed of 100 Gb/s.
- c) Data and delimiters are synchronous to a clock reference.
- d) It provides independent 64-bit wide transmit and receive data paths.
- e) It supports full duplex operation only.

81.1.1 Summary of major concepts

The following are the major concepts of the XLGMII/CGMII:

- a) The XLGMII/CGMII is functionally similar to other media independent interfaces that have been defined for lower speeds, as they all define an interface allowing independent development of MAC and PHY logic.
- b) The RS converts between the MAC serial data stream and the parallel data paths of the XLGMII/CGMII.
- c) The RS maps the signal set provided at the XLGMII/CGMII to the PLS service primitives provided at the MAC.
- d) Each direction of data transfer is independent and serviced by data, control, and clock signals.
- e) The RS generates continuous data or control characters on the transmit path and expects continuous data or control characters on the receive path.
- f) The RS participates in link fault detection and reporting by monitoring the receive path for status reports that indicate an unreliable link, and generating status reports on the transmit path to report detected link faults to the DTE on the remote end of the connecting link.
- g) The XLGMII and CGMII may also support Low Power Idle (LPI) signaling for PHY types supporting Energy Efficient Ethernet (EEE) (see Clause 78).

81.1.2 Application

This clause applies to the interface between the MAC and PHY. This logical interface is used to provide media independence so that an identical media access controller may be used with supported PHY types.

81.1.3 Rate of operation

The XLGMII is specified to support 40 Gb/s operation and the CGMII is specified to support 100 Gb/s operation.

81.1.4 Delay constraints

The maximum cumulative MAC Control, MAC, and RS delay (sum of transmit and receive delays at one end of the link) shall meet the values specified in Table 81–1. A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 80.4 and its references.

81.1.5 Allocation of functions

The allocation of functions at the XLGMII/CGMII balances the need for media independence with interface simplicity. The XLGMII and CGMII maximize media independence by separating the Data Link and Physical Layers of the OSI seven-layer reference model.

Table 81–1—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
40 Gb/s MAC, RS, and MAC Control	16384	32	409.6
100 Gb/s MAC, RS, and MAC Control	24576	48	245.76

81.1.6 XLGMII/CGMII structure

The XLGMII/CGMII is composed of independent transmit and receive paths. Each direction uses 64 data signals (TXD<63:0> and RXD<63:0>), 8 control signals (TXC<7:0> and RXC<7:0>), and a clock (TX_CLK and RX_CLK). Figure 81–2 depicts a schematic view of the RS inputs and outputs.

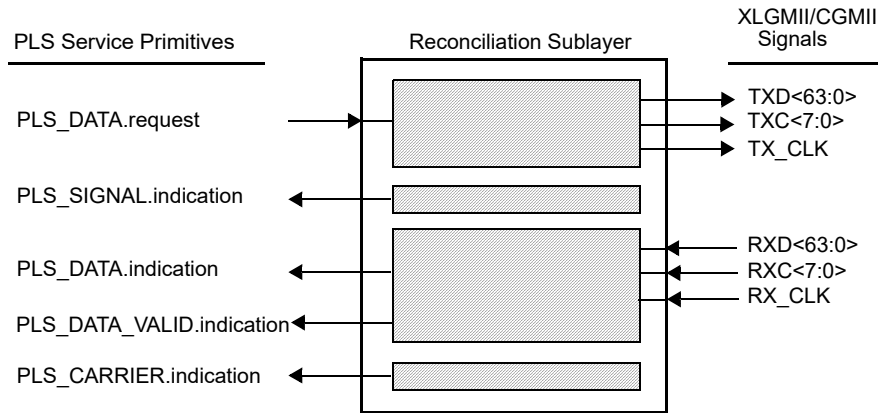


Figure 81–2—Reconciliation Sublayer (RS) inputs and outputs

The 64 TXD and 8 TXC signals shall be organized into eight data lanes, as shall the 64 RXD and 8 RXC signals (see Table 81–2). The eight lanes in each direction share a common clock, TX_CLK for transmit and RX_CLK for receive. The eight lanes are used in round-robin sequence to carry an octet stream. On transmit, each eight PLS_DATA.request transactions represent an octet transmitted by the MAC. The first octet is aligned to lane 0, the second to lane 1, the third to lane 2, the fourth to lane 3, the fifth to lane 4, the sixth to lane 5, the seventh to lane 6 and the eighth to lane 7, then repeating with the ninth to lane 0, etc. Delimiters and interframe idle characters are encoded on the TXD and RXD signals with the control code indicated by assertion of TXC and RXC, respectively.

81.1.7 Mapping of XLGMII/CGMII signals to PLS service primitives

The Reconciliation Sublayer (RS) shall map the signals provided at the XLGMII/CGMII to the PLS service primitives defined in Clause 6. The PLS service primitives provided by the RS and described here behave in exactly the same manner as defined in Clause 6. Full duplex operation only is implemented at 40 Gb/s and 100 Gb/s; therefore, PLS service primitives supporting CSMA/CD operation are not mapped through the RS to the XLGMII/CGMII. This behavior and restrictions are the same as described in 22.7, with the details of the signaling described in 81.3. LPI_REQUEST shall not be set to ASSERT unless the attached link has been operational for at least one second (i.e., link_status = OK, according to the underlying PCS/PMA).

EEE capability requires the use of the MAC defined in Annex 4A for simplified full duplex operation (with carrier sense deferral). This provides full duplex operation but uses the carrier sense signal to defer transmission when the PHY is in its low power state.

Table 81–2—Transmit and receive lane associations

TXD, RXD	TXC, RXC	Lane
<7:0>	<0>	0
<15:8>	<1>	1
<23:16>	<2>	2
<31:24>	<3>	3
<39:32>	<4>	4
<47:40>	<5>	5
<55:48>	<6>	6
<63:56>	<7>	7

Mappings for the following primitives are defined for 40 Gb/s and 100 Gb/s operation:

PLS_DATA.request
PLS_DATA.indication
PLS_CARRIER.indication
PLS_SIGNAL.indication
PLS_DATA_VALID.indication

81.1.7.1 Mapping of PLS_DATA.request

81.1.7.1.1 Function

The RS maps the primitive PLS_DATA.request to the XLGMII/CGMII signals TXD<63:0>, TXC<7:0>, and TX_CLK.

81.1.7.1.2 Semantics of the service primitive

PLS_DATA.request(OUTPUT_UNIT)

The OUTPUT_UNIT parameter can take one of three values: ONE, ZERO, or DATA_COMPLETE. ONE or ZERO represents a single data bit. The DATA_COMPLETE value signifies that the MAC sublayer has no more data to output.

81.1.7.1.3 When generated

This primitive is generated by the MAC sublayer to request the transmission of a single data bit on the physical medium or to stop transmission.

81.1.7.1.4 Effect of receipt

The OUTPUT_UNIT values are conveyed to the PHY by the signals TXD<63:0> and TXC<7:0> on each TX_CLK rising edge. Each PLS_DATA.request transaction shall be mapped to a TXD signal in sequence (TXD<0>, TXD<1>, ... TXD<63>, TXD<0>) as described in 81.2, and for every eight transactions, a TXC signal in sequence (TXC<0>, TXC<1>, ... TXC<7>, TXC<0>) is generated, as described in 81.3.1.2. After 64 PLS_DATA.request transactions from the MAC sublayer (eight octets of eight PLS_DATA.request transactions each), the RS requests transmission of 64 data bits by the PHY. The first octet of preamble shall

be converted to a Start control character and aligned to lane 0. The TXD<63:0> and TXC<7:0> shall be generated by the RS every 64 bit-times of the MAC sublayer.

The DATA_COMPLETE value shall be mapped to a Terminate control character encoded on the next eight TXD signals in sequence after the last data octet; and is transferred to the PHY at the next TX_CLK rising edge. This may be on the same TX_CLK rising edge as the last data octet or the subsequent TX_CLK rising edge. When the Terminate control character is in lane 0, 1, 2, 3, 4, 5, or 6, the lanes following in sequence are encoded with an Idle control character.

81.1.7.2 Mapping of PLS_DATA.indication

81.1.7.2.1 Function

The RS maps the XLGMII/CGMII signals RXD<63:0>, RXC<7:0>, and RX_CLK to the primitive PLS_DATA.indication.

81.1.7.2.2 Semantics of the service primitive

PLS_DATA.indication (INPUT_UNIT)

The INPUT_UNIT parameter can take one of two values: one or zero. It represents a single data bit.

81.1.7.2.3 When generated

The INPUT_UNIT values are derived from the signals RXC<7:0> and RXD<63:0> received from the PHY on each rising edge of RX_CLK. Each primitive generated to the MAC sublayer entity corresponds to a PLS_DATA.request issued by the MAC at the remote end of the link connecting two DTEs. For each RXD<63:0> during frame reception, the RS shall generate 64 PLS_DATA.indication transactions until the end of frame (Terminate control character), where 0, 8, 16, 24, 32, 40, 48, or 56 PLS_DATA.indication transactions will be generated from the RXD<63:0> containing the Terminate. During frame reception, each RXD signal shall be mapped in sequence into a PLS_DATA.indication transaction (RXD<0>, RXD<1>, ... RXD<63>, RXD<0>) as described in 81.2.

The RS shall convert a valid Start control character to a preamble octet prior to generation of the associated PLS_DATA.indication transactions. The RS shall not generate any PLS_DATA.indication primitives for a Terminate control character. To assure robust operation, the value of the data transferred to the MAC may be changed by the RS as required by XLGMII/CGMII error indications (see 81.3.3). Sequence ordered sets are not indicated to the MAC (see 81.3.4).

81.1.7.2.4 Effect of receipt

The effect of receipt of this primitive by the MAC sublayer is unspecified.

81.1.7.3 Mapping of PLS_CARRIER.indication

40 Gb/s and 100 Gb/s operation supports full duplex operation only. The RS only generates this primitive when optional EEE capability or the optional detection of Link Interruption is supported.

CARRIER_STATUS is set to CARRIER_ON if the optional EEE capability is supported and LPI_CARRIER_STATUS is TRUE, or if optional detection of Link Interruption is supported and link_fault is Link Interruption (see 81.3.4.1). CARRIER_STATUS is otherwise set to CARRIER_OFF. The deferral mechanism based upon the Link Interruption signal may be enabled or disabled by management.

81.1.7.4 Mapping of PLS_SIGNAL.indication

40 Gb/s and 100 Gb/s operation supports full duplex operation only. The RS never generates this primitive.

81.1.7.5 Mapping of PLS_DATA_VALID.indication

81.1.7.5.1 Function

The RS maps the XLGMII/CGMII signals RXC<7:0> and RXD<63:0> to the primitive PLS_DATA_VALID.indication.

81.1.7.5.2 Semantics of the service primitive

PLS_DATA_VALID.indication (DATA_VALID_STATUS)

The DATA_VALID_STATUS parameter can take one of two values: DATA_VALID or DATA_NOT_VALID. The DATA_VALID value indicates that the INPUT_UNIT parameter of the PLS_DATA.indication primitive contains valid data of an incoming frame. The DATA_NOT_VALID value indicates that the INPUT_UNIT parameter of the PLS_DATA.indication primitive does not contain valid data of an incoming frame.

81.1.7.5.3 When generated

The PLS_DATA_VALID.indication service primitive shall be generated by the RS whenever the DATA_VALID_STATUS parameter changes from DATA_VALID to DATA_NOT_VALID or vice versa.

DATA_VALID_STATUS shall assume the value DATA_VALID when a PLS_DATA.indication transaction is generated in response to reception of a Start control character on lane 0 if the prior RXC<7:0> and RXD<63:0> contained eight Idle characters, a Sequence ordered set, or a Terminate character. DATA_VALID_STATUS shall assume the value DATA_NOT_VALID when RXC of the current lane in sequence is asserted for anything except an Error control character. In the absence of errors, DATA_NOT_VALID is caused by a Terminate control character. When DATA_VALID_STATUS changes from DATA_VALID to DATA_NOT_VALID because of a control character other than Terminate, the RS shall ensure that the MAC will detect a FrameCheckError prior to indicating DATA_NOT_VALID to the MAC (see 81.3.3.1).

81.1.7.5.4 Effect of receipt

The effect of receipt of this primitive by the MAC sublayer is unspecified.

81.2 XLGMII/CGMII data stream

Packets transmitted through the XLGMII/CGMII shall be transferred within the XLGMII/CGMII data stream. A data stream is a sequence of bytes, where each byte conveys either a data octet or control character. The parts of the data stream are shown in Figure 81–3.

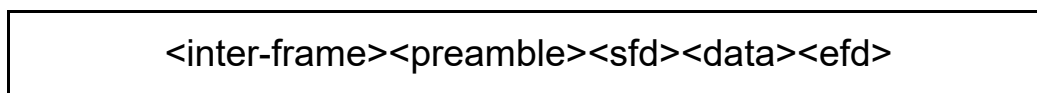


Figure 81–3—XLGMII/CGMII data stream

For the XLGMII/CGMII, transmission and reception of each bit and mapping of data octets to lanes shall be as shown in Figure 81–4.

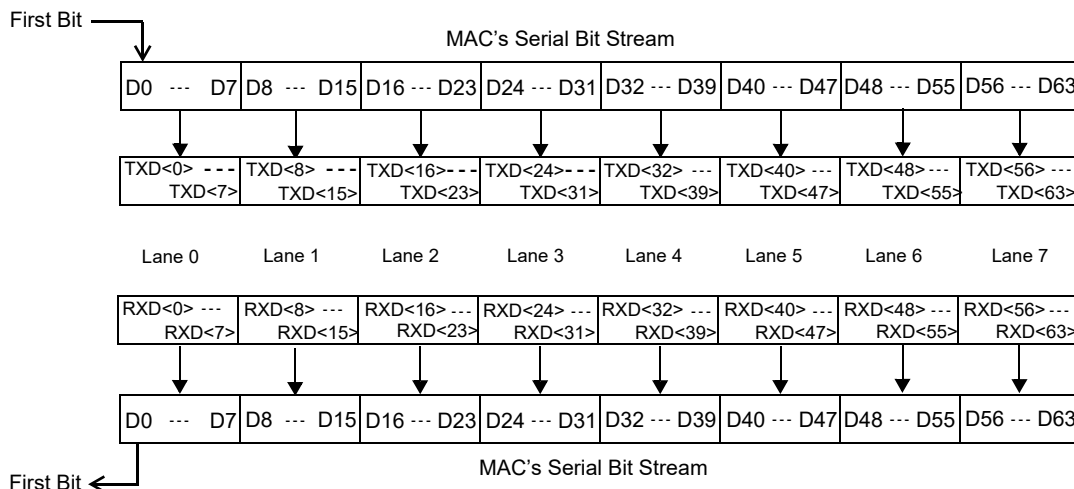


Figure 81–4—Relationship of data lanes to MAC serial bit stream

81.2.1 Inter-frame <inter-frame>

The inter-frame <inter-frame> period on an XLGMII/CGMII transmit or receive path is an interval during which no frame data activity occurs. The <inter-frame> corresponding to the MAC interpacket gap begins with the Terminate control character, continues with Idle control characters and ends with the Idle control character prior to a Start control character. The length of the interpacket gap may be changed between the transmitting MAC and receiving MAC by one or more functions (e.g., RS lane alignment or PHY clock rate compensation). The minimum IPG at the XLGMII/CGMII of the receiving RS is one octet.

The signaling of link status information logically occurs in the <inter-frame> period (see 81.3.4). Frame processing when signaling of link status information is initiated or terminated is described in 81.3.3.

81.2.2 Preamble <preamble> and start of frame delimiter <sfd>

The preamble <preamble> begins a frame transmission by a MAC as specified in 4.2.5 and when generated by a MAC consists of 7 octets with the following bit values:

10101010 10101010 10101010 10101010 10101010 10101010 10101010

The Start control character indicates the beginning of MAC data on the XLGMII/CGMII. On transmit, the RS converts the first data octet of preamble transferred from the MAC into a Start control character. On receive, the RS converts the Start control character into a preamble data octet. The Start control character is aligned to lane 0 of the XLGMII/CGMII by the RS on transmit and by the PHY on receive.

The start of frame delimiter <sfd> indicates the start of a frame and immediately follows the preamble. The bit value of <sfd> at the XLGMII/CGMII is the same as the Start Frame Delimiter (SFD) specified in 4.2.6 and is equal to the following:

10101011

The preamble and SFD are shown previously with their bits ordered for serial transmission from left to right. As shown, the left-most bit of each octet is the LSB of the octet and the right-most bit of each octet is the MSB of the octet.

The preamble and SFD are transmitted through the XLGMII/CGMII as octets sequentially ordered on the lanes of the XLGMII/CGMII. The first preamble octet is replaced with a Start control character and it is aligned to lane 0, the second octet on lane 1, the third on lane 2, the fourth on lane 3, the fifth on lane 4, the sixth on lane 5, the seventh on lane 6, and the SFD on lane 7, and the eight octets are transferred on the next rising edge of TX_CLK. The ninth octet is assigned to lane 0 with subsequent octets sequentially assigned to the lanes. The XLGMII/CGMII <preamble> and <sfd> are as follows:

Lane 0	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7
Start	10101010	10101010	10101010	10101010	10101010	10101010	10101011

81.2.3 Data <data>

The data <data> in a frame shall consist of a set of data octets.

81.2.4 End of frame delimiter <efd>

Assertion of TXC with the appropriate Terminate control character encoding of TXD on a lane constitutes an end of frame delimiter <efd> for the transmit data stream. Similarly, assertion of RXC with the appropriate Terminate control character encoding of RXD constitutes an end of frame delimiter for the receive data stream. The XLGMII/CGMII shall recognize the end of frame delimiter on any of the eight lanes of the XLGMII/CGMII.

81.2.5 Definition of Start of Packet and End of Packet Delimiters

For the purposes of Clause 30, the Start of Packet Delimiter is defined as the Start control character, and the End of Packet delimiter is defined as the end of the last sequential data octet preceding the Terminate control character or other control character causing a change from DATA_VALID to DATA_NOT_VALID (see 81.1.7.5.2 and 30.3.2.1.5).

81.3 XLGMII/CGMII functional specifications

The XLGMII/CGMII is designed to make the differences among the various media and transceiver combinations transparent to the MAC sublayer. The selection of logical control signals and the functional procedures are all designed to this end.

NOTE—No XLGMII/CGMII loopback is defined, but XLGMII/CGMII signals are specified such that transmit signals may be connected to receive signals to create a loopback path. To do this, TXD<0> is connected to RXD<0> ... TXD<63> to RXD<63>, TXC<0> to RXC<0> ... TXC<7> to RXC<7>, and TXCLK to RXCLK. Such a loopback does not test the Link Fault Signaling state diagram, nor any of the error handling functions of the receive RS.

81.3.1 Transmit

81.3.1.1 TX_CLK

TX_CLK is a continuous clock used for operation at the appropriate frequency. TX_CLK provides the timing reference for the transfer of the TXC<7:0> and TXD<63:0> signals from the RS to the PHY. The values of TXC<7:0> and TXD<63:0> shall be sampled by the PHY on the rising edge of TX_CLK. TX_CLK is sourced by the RS.

The TX_CLK frequency shall be one-sixty-fourth of the MAC transmit data rate.

81.3.1.2 TXC<7:0> (transmit control)

TXC<7:0> indicate that the RS is presenting either data or control characters on the XLGMII/CGMII for transmission. The TXC signal for a lane shall be deasserted when a data octet is being sent on the corresponding lane and asserted when a control character is being sent. In the absence of errors, the TXC signals are deasserted by the RS for each octet of the preamble (except the first octet that is replaced with a Start control character) and remain deasserted while all octets to be transmitted are presented on the lanes of the XLGMII/CGMII. TXC<7:0> are driven by the RS and shall transition synchronously with respect to the rising edge of TX_CLK. Table 81–3 specifies the permissible encodings of TXD and TXC for an XLGMII/CGMII transmit lane. Additional requirements apply for proper code sequences and in which lanes particular codes are valid (e.g., Start control character is to be aligned to lane 0).

Table 81–3—Permissible encodings of TXC and TXD

TXC	TXD	Description	PLS_DATA.request parameter
0	0x00 through 0xFF	Normal data transmission	Zero, one (eight bits)
1	0x00 through 0x05	Reserved	—
1	0x06	Only valid on all 8 lanes simultaneously to request LPI	No applicable parameter (normal inter-frame)
1	0x07	Idle	No applicable parameter (normal inter-frame)
1	0x08 through 0x9B	Reserved	—
1	0x9C	Sequence (only valid in lane 0)	No applicable parameter (inter-frame status signal)
1	0x9D through 0xFA	Reserved	—
1	0xFB	Start (only valid in lane 0)	No applicable parameter, replaces first eight zero, one of a frame (preamble octet)
1	0xFC	Reserved	—
1	0xFD	Terminate	DATA_COMPLETE
1	0xFE	Error	No applicable parameter
1	0xFF	Reserved	—

A PHY with EEE capability shall interpret the combination of TXC and TXD as shown in Table 81–3 as an assertion of LPI. Transition into and out of the LPI state is shown in Figure 81–7.

81.3.1.3 TXD<63:0> (transmit data)

TXD is a bundle of 64 data signals organized into eight lanes of eight signals each (TXD<7:0>, TXD<15:8>, TXD<23:16>, TXD<31:24>, TXD<39:32>, TXD<47:40>, TXD<55:48>, and TXD<63:56>) that are driven by the RS. Each lane is associated with a TXC signal as shown in Table 81–2 and shall be encoded as shown in Table 81–3. TXD<63:0> shall transition synchronously with respect to the rising edge of TX_CLK. For each high TX_CLK transition, data and/or control are presented on TXD<63:0> to the PHY for transmission. TXD<0> is the least significant bit of lane 0, TXD<8> the least significant bit of lane 1, TXD<16> the least significant bit of lane 2, TXD<24> the least significant bit of lane 3, TXD<32> the least significant bit of lane 4, TXD<40> the least significant bit of lane 5, TXD<48> the least significant bit of lane 6, and TXD<56> the least significant bit of lane 7.

Assertion on a lane of appropriate TXD values when TXC is asserted will cause the PHY to generate code-groups associated with either Idle, Start, Terminate, Sequence, or Error control characters. While the TXC of a lane is deasserted, TXD of the lane is used to request the PHY to generate code-groups corresponding to the data octet value of TXD. An example of normal frame transmission is illustrated in Figure 81–5.

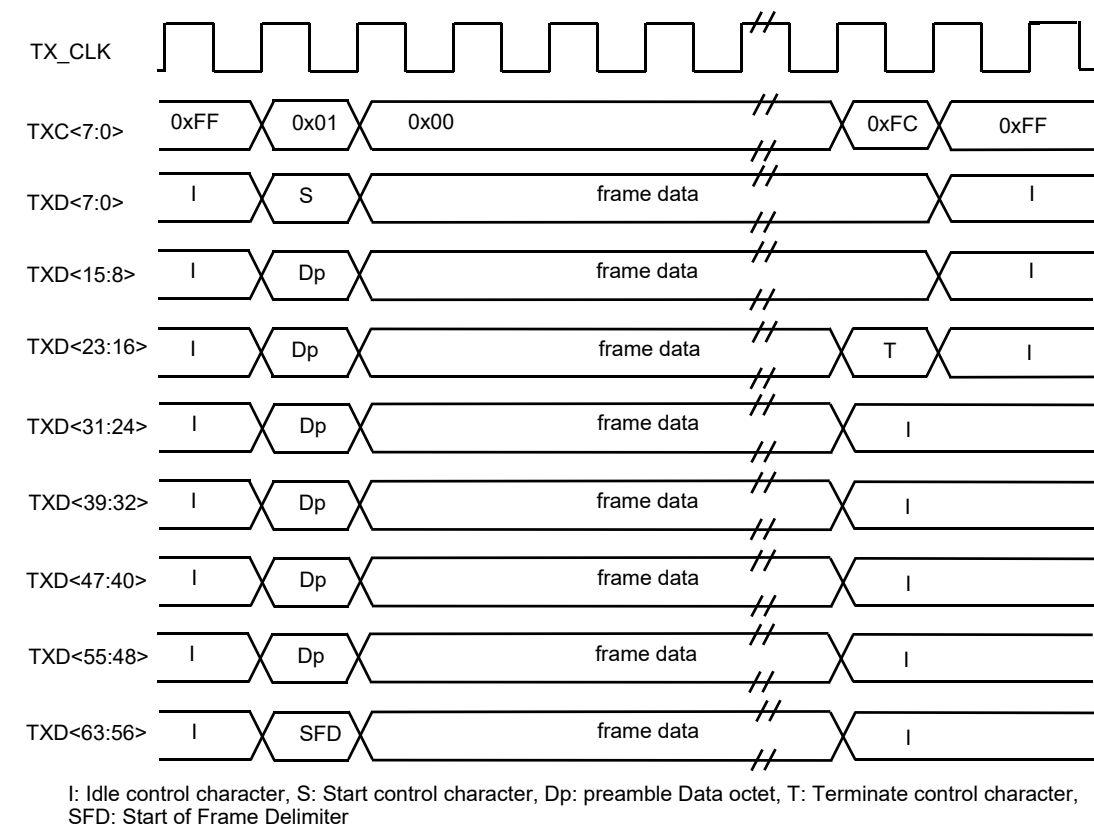
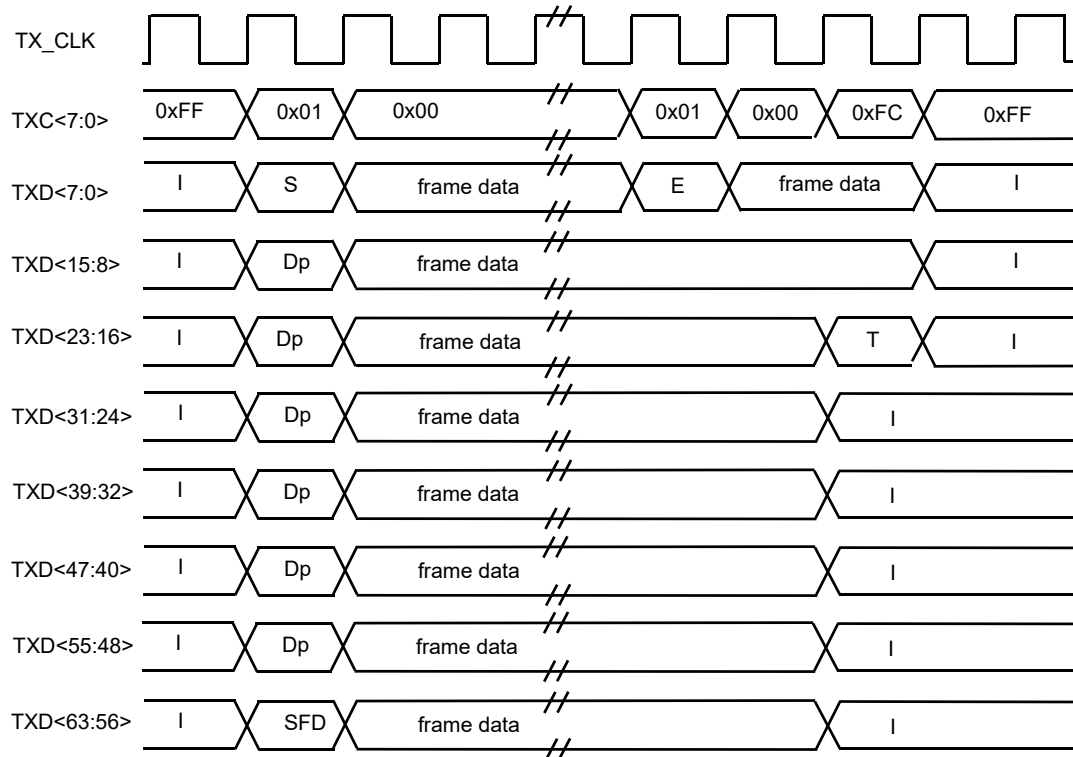


Figure 81–5—Normal frame transmission

Figure 81–6 shows the behavior of TXD and TXC during an example transmission of a frame propagating an error.



I: Idle control character, S: Start control character, Dp: preamble Data octet, T: Terminate control character, E: Error control character, SFD: Start of Frame Delimiter

Figure 81–6—Transmit Error Propagation

81.3.1.4 Start control character alignment

On transmit, it may be necessary for the RS to modify the length of the <inter-frame> in order to align the Start control character (first octet of preamble) on lane 0. This shall be accomplished in one of the following two ways:

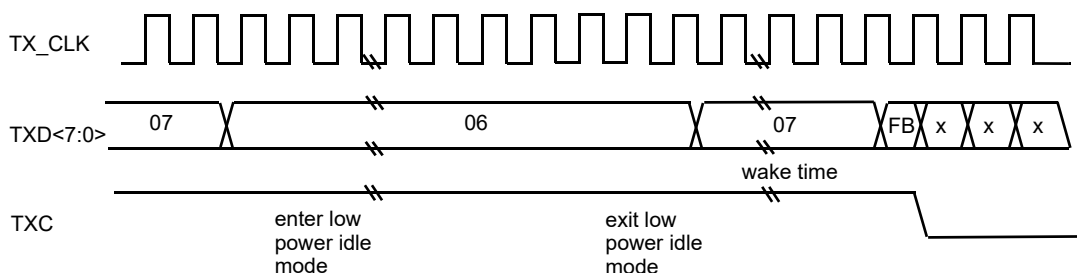
- A MAC implementation may incorporate this RS function into its design and always insert additional idle characters to align the start of preamble on an eight byte boundary. Note that this will reduce the effective data rate for certain packet sizes separated with minimum inter-frame spacing.
- Alternatively, the RS may maintain the effective data rate by sometimes inserting and sometimes deleting idle characters to align the Start control character. When using this method the RS maintains a Deficit Idle Count (DIC) that represents the cumulative count of idle characters deleted or inserted. The DIC is incremented for each idle character deleted, decremented for each idle character inserted, and the decision of whether to insert or delete idle characters is constrained by bounding the DIC to a minimum value of zero and maximum value of seven. Note that this may result in inter-frame spacing observed on the transmit XLGMII/CGMII that is up to seven octets shorter than the minimum transmitted inter-frame spacing specified in Clause 4; however, the frequency of shortened inter-frame spacing is constrained by the DIC rules. The DIC is only reset at initialization and is applied regardless of the size of the IPG transmitted by the MAC sublayer. An equivalent technique may be employed to control RS alignment of the Start control character provided that the result is the same as if the RS implemented the DIC as described.

81.3.1.5 Transmit direction LPI transition

LPI operation and the LPI client are described in 78.1. The RS requests the PHY to transition to the LPI state by asserting TXC and setting TXD to 0x06 (in all lanes). The RS maintains the same state for these signals for the entire time that the PHY is to remain in the LPI state.

The RS asserts TXC and asserts IDLE on lanes 0 to 7 in order to make the PHY transition out of the LPI state. The RS should not present a start code for valid transmit data until after the wake-up time specified for the PHY ($T_{w_sys_tx}$). The wake times are shown in Table 78–4

Figure 81–7 shows the behavior of TXC and TXD<7:0> during the transition into and out of the LPI state.



NOTE—TXC and TXD are shown for one lane, all 8 lanes behave identically during LPI.

Figure 81–7—LPI transition

Table 81–3 summarizes the permissible encodings of TXD<63:0>, TXC<7:0>.

81.3.2 Receive

81.3.2.1 RX_CLK (receive clock)

RX_CLK is a continuous clock that provides the timing reference for the transfer of the RXC<7:0> and RXD<63:0> signals from the PHY to the RS. RXC<7:0> and RXD<63:0> shall be sampled by the RS on the rising edge of RX_CLK. RX_CLK is sourced by the PHY.

The frequency of RX_CLK may be derived from the received data or it may be that of a nominal clock (e.g., TX_CLK). When the received data rate at the PHY is within tolerance, the RX_CLK frequency shall be one-sixty-fourth of the MAC receive data rate.

There is no need to transition between the recovered clock reference and a nominal clock reference on a frame-by-frame basis. If loss of received signal from the medium causes a PHY to lose the recovered RX_CLK reference, the PHY shall source the RX_CLK from a nominal clock reference.

NOTE—This standard neither requires nor assumes a guaranteed phase relationship between the RX_CLK and TX_CLK signals.

81.3.2.2 RXC<7:0> (receive control)

RXC<7:0> indicate that the PHY is presenting either recovered and decoded data or control characters on the XLGMII/CGMII. The RXC signal for a lane shall be deasserted when a data octet is being received on the corresponding lane and asserted when a control character is being received. In the absence of errors, the RXC signals are deasserted by the PHY for each octet of the preamble (except the first octet that is replaced with a Start control character) and remain deasserted while all octets to be received are presented on the

lanes of the XLGMII/CGMII. $RXC_{<7:0>}$ are driven by the PHY and shall transition synchronously with respect to the rising edge of RX_CLK . Table 81–4 specifies the permissible encodings of RXD and RXC for an XLGMII/CGMII receive lane. Additional requirements apply for proper code sequences and in which lanes particular codes are valid (e.g., Start control character is to be aligned to lane 0).

Table 81–4—Permissible lane encodings of RXD and RXC

RXC	RXD	Description	PLS_DATA.indication parameter
0	0x00 through 0xFF	Normal data reception	Zero, one (eight bits)
1	0x00 through 0x05	Reserved	—
1	0x06	Only valid on all 8 lanes simultaneously to indicate LP_IDLE is asserted	No applicable parameter (normal inter-frame)
1	0x07	Idle	No applicable parameter (Normal inter-frame)
1	0x08 through 0x9B	Reserved	—
1	0x9C	Sequence (only valid in lane 0)	No applicable parameter (Inter-frame status signal)
1	0x9D through 0xFA	Reserved	—
1	0xFB	Start (only valid in lane 0)	No applicable parameter, first eight zero, one of a frame (a preamble octet)
1	0xFC	Reserved	—
1	0xFD	Terminate	No applicable parameter (start of inter-frame)
1	0xFE	Error	No applicable parameter
1	0xFF	Reserved	—

Figure 81–8 shows the behavior of RXC<7:0> during an example frame reception with no errors.

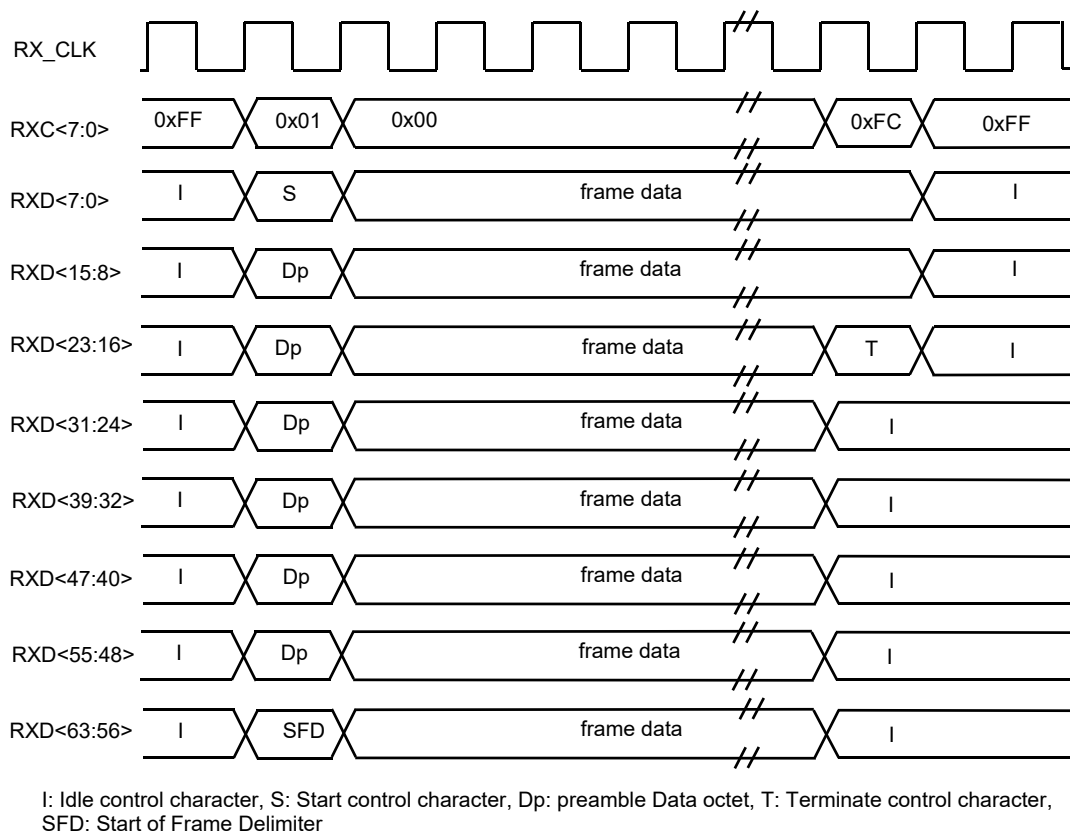


Figure 81–8—Frame reception without error

81.3.2.3 RXD (receive data)

RXD is a bundle of 64 data signals (RXD<63:0>) organized into eight lanes of eight signals each (RXD<7:0>, RXD<15:8>, RXD<23:16>, RXD<31:24>, RXD<39:32>, RXD<47:40>, RXD<55:48>, and RXD<63:56>) that are driven by the PHY. Each lane is associated with an RXC signal as shown in Table 81–2 and shall be decoded by the RS as shown in Table 81–4. RXD<63:0> shall transition synchronously with respect to the rising edge of RX_CLK. For each rising RX_CLK transition, received data and/or control are presented on RXD<63:0> for mapping by the RS. RXD<0> is the least significant bit of lane 0, RXD<8> the least significant bit of lane 1, RXD<16> the least significant bit of lane 2, RXD<24> the least significant bit of lane 3, RXD<32> the least significant bit of lane 4, RXD<40> the least significant bit of lane 5, RXD<48> the least significant bit of lane 6, and RXD<56> the least significant bit of lane 7. Figure 81–8 shows the behavior of RXD<63:0> during frame reception.

While the RXC of a lane is deasserted, RXD of the lane is used by the RS to generate PLS_DATA indications. Assertion on a lane of appropriate RXD values when RXC is asserted indicates to the RS the Start control character, Terminate control character, Sequence control character, or Error control character that drive its mapping functions.

RXC of a lane is asserted with the appropriate Error control character encoding on RXD of the lane to indicate that an error was detected somewhere in the frame presently being transferred from the PHY to the

RS (e.g., a coding error, or any error that the PHY is capable of detecting, and that may otherwise be undetectable at the MAC sublayer).

The effect of an Error control character on the RS is defined in 81.3.3.1. Figure 81–9 shows the behavior of RXC and RXD during the reception of an example frame with an error.

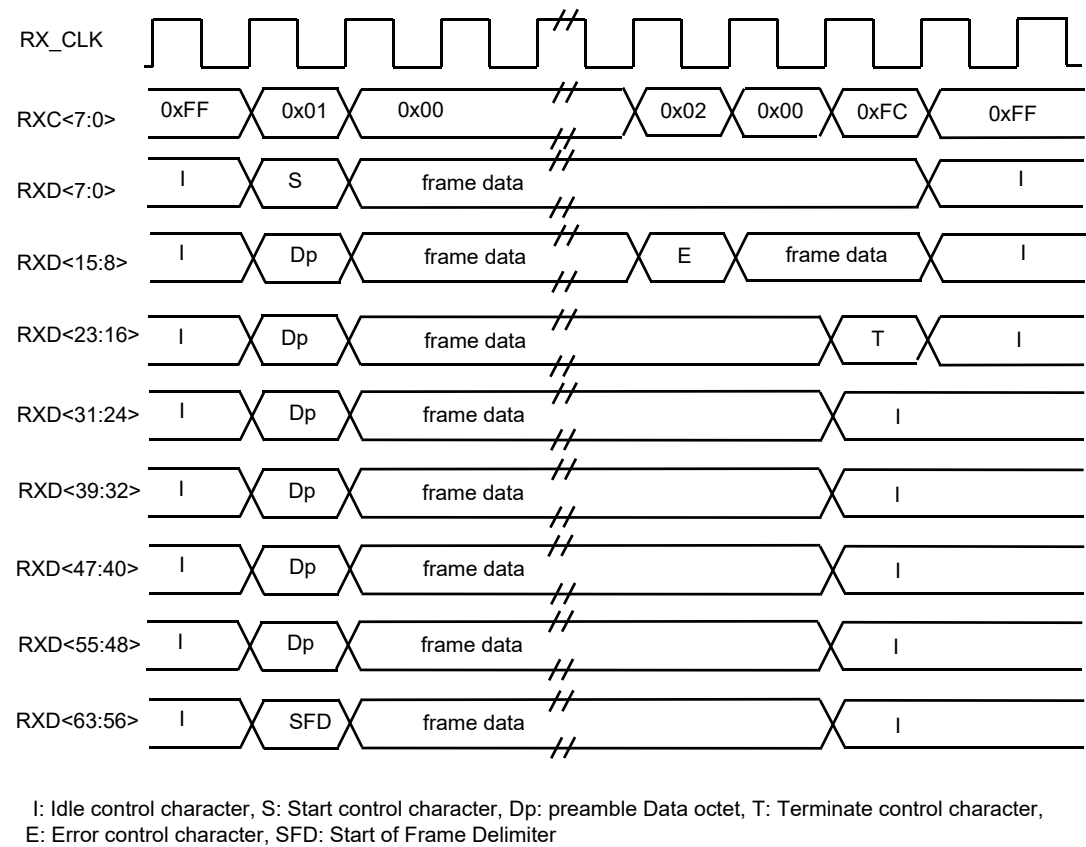
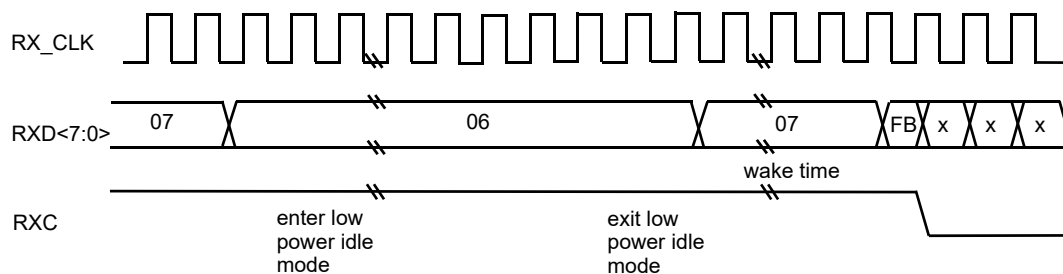


Figure 81–9—Reception with error

81.3.2.4 Receive direction LPI transition

LPI operation and the LPI client are described in 78.1. When the PHY receives signals from the link partner to indicate transition into the low power state, it indicates this to the RS by asserting RXC and setting RXD to 0x06 (in all lanes). The PHY maintains these signals in this state while it remains in the LPI state. When the PHY receives signals from the link partner to indicate transition out of the LPI state, it indicates this to the RS by asserting RXC and asserting idle on all lanes 0 to 7 to return to a normal interframe state. The RS shall interpret the LPI coding as shown in Table 81–4.

Figure 81–10 shows the behavior of RXC and RXD<7:0> during LPI transitions.



NOTE 1—RXC and RXD are shown for one lane, all 8 lanes behave identically during LPI.

NOTE 2—In some instances, LPI may be followed by characters other than IDLE during wake time.

Figure 81–10—LPI transition

81.3.3 Error and fault handling

81.3.3.1 Response to error indications by the XLGMII/CGMII

If, during frame reception (i.e., when DATA_VALID_STATUS = DATA_VALID), a control character other than a Terminate control character is signaled on a received lane, the RS shall ensure that the MAC will detect a FrameCheckError in that frame. This requirement may be met by incorporating a function in the RS that produces a received frame data sequence delivered to the MAC sublayer that is guaranteed to not yield a valid CRC result, as specified by the frame check sequence algorithm (see 3.2.8). This data sequence may be produced by substituting data delivered to the MAC. The RS generates eight PLS_DATA.indication primitives for each Error control character received within a frame, and may generate eight PLS_DATA.indication primitives to ensure FrameCheckError when a control character other than Terminate causes the end of the frame.

Other techniques may be employed to respond to a received Error control character provided that the result is that the MAC sublayer behaves as though a FrameCheckError occurred in the received frame.

81.3.3.2 Conditions for generation of transmit Error control characters

If, during the process of transmitting a frame, it is necessary to request that the PHY deliberately corrupt the contents of the frame in such a manner that a receiver will detect the corruption with the highest degree of probability, then an Error control character may be asserted on a transmit lane by the appropriate encoding of the lane's TXD and TXC signals.

81.3.3.3 Response to received invalid frame sequences

The RS shall not indicate DATA_VALID to the MAC for a Start control character received on any lane other than lane 0. Error free operation will not change the SFD alignment in lane 7. A MAC/RS implementation is not required to process a packet that has an SFD in a position other than lane 7 in the XLGMII/CGMII transfer containing the Start control character.

81.3.4 Link fault signaling

Link fault signaling operates between the remote RS and the local RS. Faults detected between the remote RS and the local RS are received by the local RS as Local Fault. Only an RS originates Remote Fault signals. The behavior of the fault signaling is the same as it is for Clause 46 with the exception that the ordered sets are aligned to eight byte boundaries, padding lanes 4 to 7 with 0x00.

Clause 46 uses the term column when describing data transfers on the XGMII. The eight lanes of data and control transferred per clock cycle on XLGMII/CGMII are equivalent to a column in the following description of link fault signaling.

Sublayers within the PHY are capable of detecting faults that render a link unreliable for communication. Upon recognition of a fault condition, a PHY sublayer indicates Local Fault status on the data path. When this Local Fault status reaches an RS, the RS stops sending MAC data or LPI, and continuously generates a Remote Fault status on the transmit data path (possibly truncating a MAC frame being transmitted). When Remote Fault status is received by an RS, the RS stops sending MAC data or LPI, and continuously generates Idle control characters. When the RS no longer receives fault status messages, it returns to normal operation, sending MAC data or LPI. Note that this behavior only supports bidirectional operation.

Status is signaled in an eight byte Sequence ordered set as shown in Table 81–5. The PHY indicates Local Fault with a Sequence control character in lane 0 and data characters of 0x00 in lanes 1, 2, 4, 5, 6, and 7 plus a data character of 0x01 in lane 3. The RS indicates a Remote Fault with a Sequence control character in lane 0 and data characters of 0x00 in lanes 1, 2, 4, 5, 6, and 7 plus a data character of 0x02 in lane 3. Though most fault detection is on the receive data path of a PHY, in some specific sublayers, faults can be detected on the transmit side of the PHY. This is also indicated by the PHY with a Local Fault status. All other values in lanes 1 to 3 not shown in Table 81–5 are reserved. The link fault signaling state diagram allows future standardization of reserved Sequence ordered sets for functions other than link fault indications.

For operation with links that may be temporarily interrupted, optional detection of a third fault condition, Link Interruption, is provided. Link Interruption is indicated by the PHY receive function by continuously sending the Link Interruption ordered set as defined in Table 81–5.

Table 81–5—Sequence ordered sets

Lane 0	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Description
Sequence	0x00	0x00	0x00	0x00	0x00	0x00	0x00	Reserved
Sequence	0x00	0x00	0x01	0x00	0x00	0x00	0x00	Local Fault
Sequence	0x00	0x00	0x02	0x00	0x00	0x00	0x00	Remote Fault
Sequence	0x00	0x00	0x03	0x00	0x00	0x00	0x00	Link Interruption

The RS reports the fault status of the link. Local Fault indicates a fault detected on the receive data path between the remote RS and the local RS. Remote Fault indicates a fault on the transmit path between the local RS and the remote RS. The RS shall implement the link fault signaling state diagram (see Figure 81–11).

81.3.4.1 Variables and counters

The Link Fault Signaling state diagram uses the following variables and counters:

col_cnt

A count of the number of columns received not containing a fault_sequence. This counter increments at RX_CLK rate (on the rising clock transitions) unless reset.

fault_sequence

A new column received on RXC<7:0> and RXD<63:0> comprising a Sequence ordered set of eight bytes and consisting of a Sequence control character in lane 0 and a seq_type in lanes 1, 2, 3, 4, 5, 6, and 7 indicating either Local Fault, Remote Fault, or Link Interruption.

last_seq_type

The seq_type of the previous Sequence ordered set received

Values: Local Fault; 0x00 in lane 1, 0x00 in lane 2, 0x01 in lane 3, 0x00 in lane 4, 0x00 in lane 5, 0x00 in lane 6, 0x00 in lane 7.

Remote Fault; 0x00 in lane 1, 0x00 in lane 2, 0x02 in lane 3, 0x00 in lane 4, 0x00 in lane 5, 0x00 in lane 6, 0x00 in lane 7.

Link Interruption; 0x00 in lane 1, 0x00 in lane 2, 0x03 in lane 3, 0x00 in lane 4, 0x00 in lane 5, 0x00 in lane 6, 0x00 in lane 7.

link_fault

An indicator of the fault status.

Values: OK; No fault.

Local Fault; fault detected by the PHY.

Remote Fault; fault detection signaled by the remote RS.

Link Interruption; link temporarily unavailable, signaled by the PHY.

reset

Condition that is true until such time as the power supply for the device that contains the RS has reached the operating region.

Values: FALSE: The device is completely powered and has not been reset (default).

TRUE: The device has not been completely powered or has been reset.

seq_cnt

A count of the number of received Sequence ordered sets of the same type.

seq_type

The value received in the current Sequence ordered set

Values: Local Fault; 0x00 in lane 1, 0x00 in lane 2, 0x01 in lane 3, 0x00 in lane 4, 0x00 in lane 5, 0x00 in lane 6, 0x00 in lane 7.

Remote Fault; 0x00 in lane 1, 0x00 in lane 2, 0x02 in lane 3, 0x00 in lane 4, 0x00 in lane 5, 0x00 in lane 6, 0x00 in lane 7.

Link Interruption; 0x00 in lane 1, 0x00 in lane 2, 0x03 in lane 3, 0x00 in lane 4, 0x00 in lane 5, 0x00 in lane 6, 0x00 in lane 7.

81.3.4.2 State diagram

The Link Fault Signaling state diagram specifies the RS monitoring of RXC<7:0> and RXD<63:0> for Sequence ordered sets. The variable link_fault is set to indicate the value of a received Sequence ordered set when four fault_sequences containing the same fault value have been received with fault sequence separated by less than 128 columns and no intervening fault_sequences of a different fault value. The state diagram is shown in Figure 81–11.

The variable link_fault is set to OK following any interval of 128 columns not containing a Remote Fault, Local Fault, or Link Interruption Sequence ordered set.

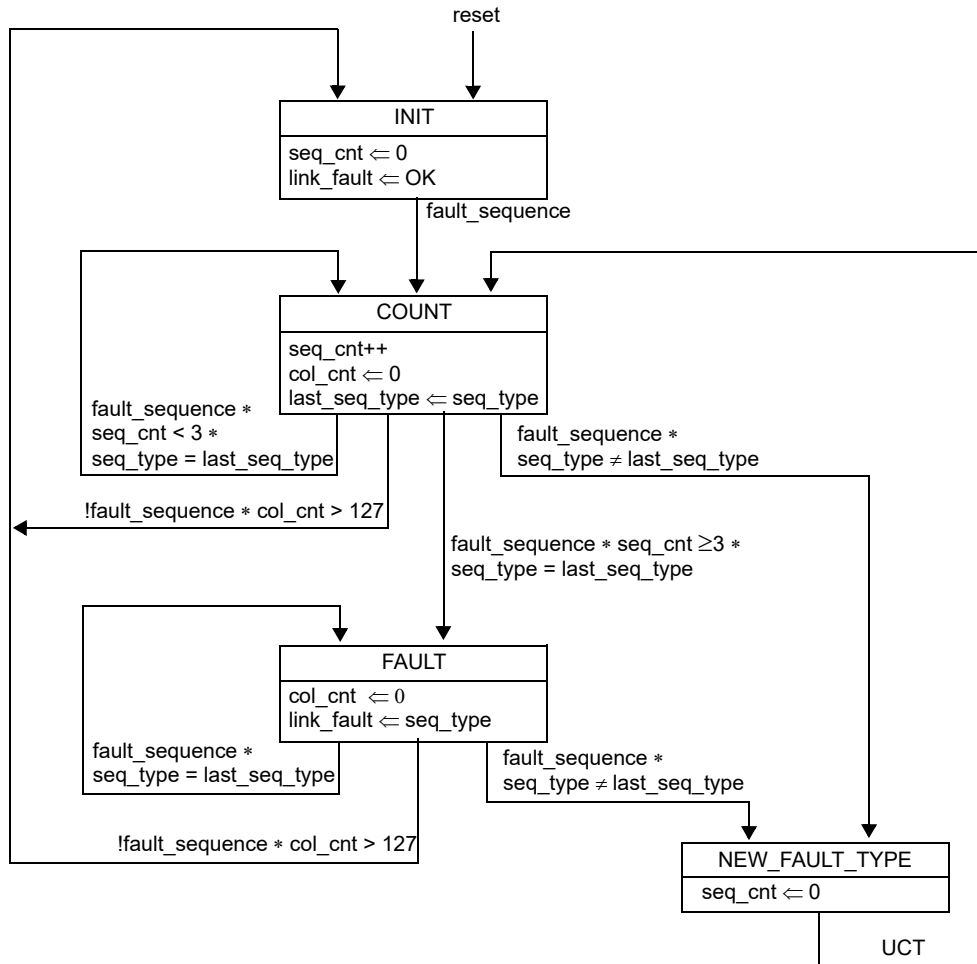


Figure 81-11—Link Fault Signaling state diagram

The RS output onto TXC<7:0> and TXD<63:0> is controlled by the variable link_fault.

- link_fault = OK
The RS shall send MAC frames as requested through the PLS service interface. In the absence of MAC frames, the RS shall generate Idle control characters.
- link_fault = Local Fault
The RS shall continuously generate Remote Fault Sequence ordered sets.
- link_fault = Remote Fault or link_fault = Link Interruption
The RS shall continuously generate Idle control characters.

81.4 LPI assertion and detection

Certain PHYs support Energy Efficient Ethernet (see Clause 78). PHYs with EEE capability support LPI assertion and detection. LPI operation and the LPI client are described in 78.1. LPI signaling allows the RS to signal to the PHY and to the link partner that a break in the data stream is expected and components may use this information to enter power saving modes that require additional time to resume normal operation. Similarly, it allows the LPI client to understand that the link partner has sent such an indication.

The LPI assertion and detection mechanism fits conceptually between the PLS Service Primitives and the XLGMII and CGMII signals as shown in Figure 81–12.

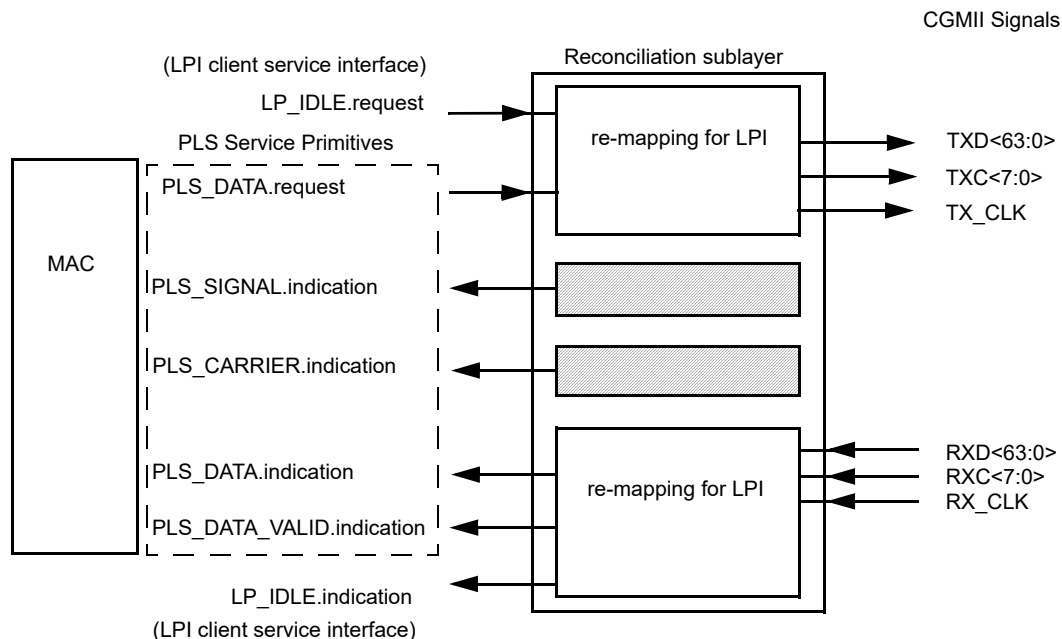


Figure 81–12—LPI assertion and detection mechanism

The definition of TXC<7:0> and TXD<63:0> is derived from the state of PLS_DATA.request (81.1.7), except when it is overridden by an assertion of Remote Fault or LP_IDLE.request.

Similarly, RXC<7:0> and RXD<63:0> are mapped to PLS_DATA.indication except when LP_IDLE is detected.

PLS_CARRIER.indication(CARRIER_STATUS) is set to CARRIER_ON when the link is in LPI mode. See 81.1.7.3.

The timing of PLS_CARRIER.indication when used for the LPI function is controlled by the LPI transmit state diagram.

81.4.1 LPI messages

LP_IDLE.indication(LPI_INDICATION)

A primitive that indicates to the LPI client that the PHY has detected the assertion or deassertion of LPI from the link partner.

Values: DEASSERT: The link partner is operating with normal inter-frame behavior (default).
ASSERT: The link partner has asserted LPI.

LP_IDLE.request(LPI_REQUEST)

The LPI_REQUEST parameter can take one of two values: ASSERT or DEASSERT. ASSERT initiates the signaling of LPI to the link partner. DEASSERT stops the signaling of LPI to the link partner. The effect of receipt of this primitive is undefined if link_status is not OK (see 73.9.1.1) or within 1 s of the change of link_status to OK.

81.4.2 Transmit LPI state diagram

The operation of LPI in the PHY requires that the MAC does not send valid data for a time after LPI has been deasserted as governed by resolved Transmit T_{w_sys} defined in 78.4.2.3.

This wake-up time is enforced by the transmit LPI state diagram using PLS_CARRIER.indication(CARRIER_STATUS). The implementation shall conform to the behavior described by the transmit LPI state diagram shown in Figure 81–13.

81.4.2.1 Variables and counters

The transmit LPI state diagram uses the following variables and counters:

LPI_CARRIER_STATUS

The LPI_CARRIER_STATUS variable indicates how the CARRIER_STATUS parameter is controlled by the LPI_REQUEST parameter. The LPI_CARRIER_STATUS is either TRUE or FALSE as determined by the Transmit LPI state diagram in Figure 81–13.

power_on

Condition that is true until such time as the power supply for the device that contains the RS has reached the operating region.

Values: FALSE: The device is completely powered (default).

TRUE: The device has not been completely powered.

reset

Used by management to control the resetting of the RS.

Values: FALSE: Do not reset the RS (default).

TRUE: Reset the RS.

tw_timer

A timer that counts the time since the deassertion of LPI. The terminal count of the timer is the value of the resolved $T_{w_sys_tx}$ as defined in 78.2. If PMA Ingress AUI Stop Enable (PIASE) bit (1.7.9) is asserted for any of the PMA sublayers, the terminal count of the timer is the value of the resolved $T_{w_sys_tx}$ as defined in 78.2 plus additional time equal to $T_{w_sys_tx} - T_{w_sys_rx}$ for the XLAUI and CAUI-n as shown in Table 78–4 for each PMA with PIASE to be asserted. The signal tw_timer_done is asserted when tw_timer reaches its terminal count.

81.4.2.2 State diagram

81.4.3 Considerations for transmit system behavior

The transmit system should expect that egress data flow is halted for at least resolved $T_{w_sys_tx}$ (see 78.2) time after it requests the deassertion of LPI. Buffering and queue management should be designed to accommodate this behavior.

81.4.4 Considerations for receive system behavior

The mapping function of the Reconciliation Sublayer shall continue to signal DATA_NOT_VALID on PLS_DATA_VALID.indication while it is detecting LP_IDLE on the XLGMII and CGMII. The receive system should be aware that data frames may arrive at the XLGMII and CGMII following the deassertion of LPI_INDICATION with a delay corresponding to the link partner's resolved $T_{w_sys_rx}$ (as specified in 78.5) time.

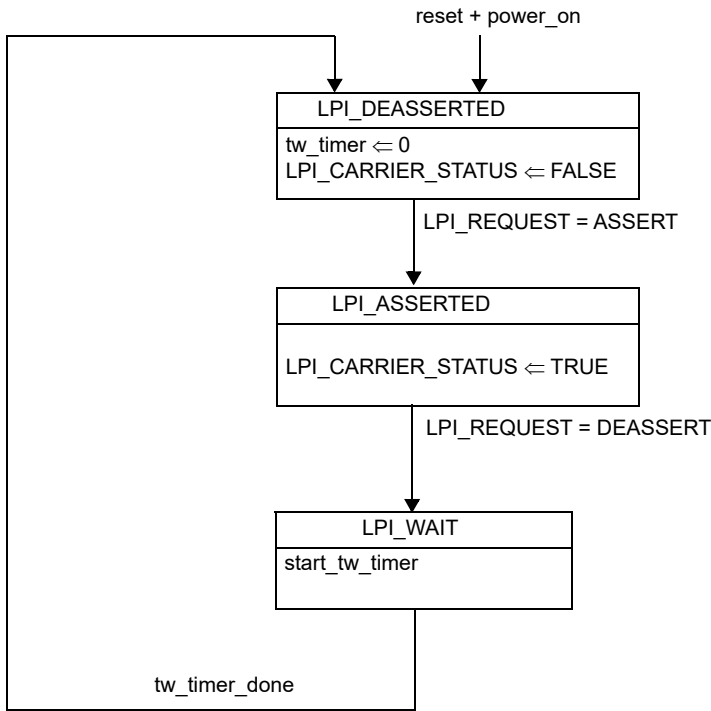


Figure 81–13—Transmit LPI State Diagram

If the PMA Ingress AUI Stop Enable (PIASE) bit (1.7.9) is asserted for any of the PMA sublayers, the PMA may stop signaling on the XLAUI and CAUI-n in the receive direction to conserve energy. The receiver should negotiate an additional time for the remote T_{w_sys} (equal to $T_{w_sys_tx} - T_{w_sys_rx}$ for the XLAUI and CAUI-n as shown in Table 78–4) for each PMA with PIASE to be asserted before setting the PIASE bits.

81.5 Protocol implementation conformance statement (PICS) proforma for Clause 81, Reconciliation Sublayer (RS) and Media Independent Interface for 40 Gb/s and 100 Gb/s operation¹⁴⁴

81.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 81, Reconciliation Sublayer (RS) and Media Independent Interface for 40 Gb/s and 100 Gb/s operation, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

81.5.2 Identification

81.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

81.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 81, Reconciliation Sublayer (RS) and Media Independent Interface for 40 Gb/s and 100 Gb/s operation
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	

Date of Statement	
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¹⁴⁴Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

81.5.2.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*PHY40	PHY support of XLGMII	81.2, 81.3		O	Yes [] No []
*PHY100	PHY support of CGMII	81.2, 81.3		O	Yes [] No []
*MII	Reconciliation Sublayer support of either XLGMII or CGMII	81.2, 81.3		O	Yes [] No []
*LPI	Implementation of LPI	81.1.7		O	Yes [] No []

81.5.3 PICS proforma tables for Reconciliation Sublayer and Media Independent Interface for 40 Gb/s and 100 Gb/s operation

81.5.3.1 General

Item	Feature	Subclause	Value/Comment	Status	Support
G1	PHY support of MAC data rate	81.1.3	Support MAC data rate of 40 Gb/s	PHY40:M	Yes [] N/A []
G2	PHY support of MAC data rate	81.1.3	Support MAC data rate of 100 Gb/s	PHY100:M	Yes [] N/A []
G3	Cumulative MAC Control, MAC and RS delay	81.1.4	Per Table 81–1	MII:M	Yes [] N/A []
G4	Lane structure	81.1.6	Per Table 81–2	M	Yes []

81.5.3.2 Mapping of PLS service primitives

Item	Feature	Subclause	Value/Comment	Status	Support
PL1	Mapping to Clause 6	81.1.7	RS implements mapping to Clause 6 PLS service primitives	MII:M	Yes [] N/A []
PL2	Mapping of PLS_DATA.requests	81.1.7.1.4	In sequence TXD<0> to TXD<63>	MII:M	Yes [] N/A []
PL3	Start control character creation	81.1.7.1.4	First octet of preamble converted to Start control character	MII:M	Yes [] N/A []
PL4	TXD and TXC generation	81.1.7.1.4	For each 64 PLS_DATA.requests	MII:M	Yes [] N/A []
PL5	Terminate control character creation	81.1.7.1.4	DATA_COMPLETE causes creation of Terminate control character in next lane in sequence	MII:M	Yes [] N/A []
PL6	Mapping RXD to PLS_DATA.indications	81.1.7.2.3	Create PLS_DATA.indications in sequence from RXD<0> to RXD<63>	MII:M	Yes [] N/A []
PL7	PLS_DATA.indication generation	81.1.7.2.3	Generate 64 PLS_DATA.indications for each RXD<63:0> until Terminate then generating 0, 8, 16, 24, 32, 40, 48, or 56	MII:M	Yes [] N/A []
PL8	Start control character conversion	81.1.7.2.3	Convert valid Start control character to preamble before generating PLS_DATA.indications	MII:M	Yes [] N/A []
PL9	Terminate control character	81.1.7.2.3	No PLS_DATA.indications generated	MII:M	Yes [] N/A []
PL10	PLS_DATA_VALID.indication generation	81.1.7.5.3	On change of value of DATA_VALID_STATUS	MII:M	Yes [] N/A []
PL11	DATA_VALID_STATUS	81.1.7.5.3	Value of DATA_VALID on a lane 0 Start control character preceded by eight idles, a Sequence ordered set, or a Terminate character	MII:M	Yes [] N/A []
PL12	DATA_VALID_STATUS	81.1.7.5.3	Value of DATA_NOT_VALID on any control character but Error	MII:M	Yes [] N/A []
PL13	Frame not ending with Terminate control character	81.1.7.5.3	Ensure MAC detects CRC error	MII:M	Yes [] N/A []

81.5.3.3 Data stream structure

Item	Feature	Subclause	Value/Comment	Status	Support
DS1	Frame transfer	81.2	Within XLGMII/CGMII data stream	MII:M	Yes [] N/A []
DS2	Bit mapping	81.2	Per Figure 81–4	MII:M	Yes [] N/A []
DS3	Content of <data>	81.2.3	Consist of data octets	MII:M	Yes [] N/A []
DS4	Recognition of <efd>	81.2.4	Terminate recognized in any lane	MII:M	Yes [] N/A []

81.5.3.4 XLGMII/CGMII signal functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
FS1	TX_CLK active edge	81.3.1.1	TXD and TXC sampled on the rising edge of TX_CLK	MII:M	Yes [] N/A []
FS2	TX_CLK frequency	81.3.1.1	One-sixty-fourth of the MAC transmit data rate	MII:M	Yes [] N/A []
FS3	TXC assertion and deassertion	81.3.1.2	Deasserted for data, asserted for control character	MII:M	Yes [] N/A []
FS4	TXC clock	81.3.1.2	Synchronous to TX_CLK	MII:M	Yes [] N/A []
FS5	TXD encoding	81.3.1.3	Per Table 81–3	MII:M	Yes [] N/A []
FS6	TXD clock	81.3.1.3	Synchronous to TX_CLK	MII:M	Yes [] N/A []
FS7	Start alignment	81.3.1.4	Start control character aligned to lane 0	MII:M	Yes [] N/A []
FS8	RX_CLK active edge	81.3.2.1	RXD and RXC sampled on the rising edge of RX_CLK	MII:M	Yes [] N/A []
FS9	RX_CLK frequency	81.3.2.1	One-sixty-fourth of the MAC receive data rate	MII:M	Yes [] N/A []
FS10	Loss of receive signal	81.3.2.1	Source RX_CLK from nominal clock	PHY:M	Yes [] N/A []
FS11	RXC assertion and deassertion	81.3.2.2	Deasserted for data, asserted for control character	PHY:M	Yes [] N/A []
FS12	RXC clock	81.3.2.2	Synchronous to RX_CLK	MII:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
FS13	RXD decoding	81.3.2.3	Per Table 81–4	MII:M	Yes [] N/A []
FS14	RXD clock	81.3.2.3	Synchronous to RX_CLK	MII:M	Yes [] N/A []
FS15	Received Error control character	81.3.3.1	RS cause MAC FrameCheck-Error	MII:M	Yes [] N/A []
FS16	DATA_VALID assertion	81.3.3.3	RS not assert DATA_VALID unless Start control character in lane 0	MII:M	Yes [] N/A []

81.5.3.5 Link fault signaling state diagram

Item	Feature	Subclause	Value/Comment	Status	Support
LF1	Link fault signaling state diagram	81.3.4	Implement per Figure 81–11	MII:M	Yes [] N/A []
LF2	link_fault = OK and MAC frames	81.3.4.2	RS services MAC frame transmission requests	MII:M	Yes [] N/A []
LF3	link_fault = OK and no MAC frames	81.3.4.2	In absence of MAC frames, RS transmits Idle control characters	MII:M	Yes [] N/A []
LF4	link_fault = Local Fault	81.3.4.2	RS transmits continuous Remote Fault Sequence ordered sets	MII:M	Yes [] N/A []
LF5	link_fault = Remote Fault	81.3.4.2	RS transmits continuous Idle control characters	MII:M	Yes [] N/A []

81.5.3.6 LPI functions

Item	Feature	Subclause	Value/Comment	Status	Support
L1	Assertion of LPI in Tx direction	81.3.1.2	As defined in Table 81–3	LPI:M	Yes [] N/A []
L2	Assertion of LPI wake time	81.4.2	As described by Figure 81–13	LPI:M	Yes [] N/A []
L3	Assertion of LPI in Rx direction	81.3.2.2	As defined in Table 81–4	LPI:M	Yes [] N/A []
L4	Signal DATA_NOT_VALID on PLS_DATA_VALID.indication	81.4.4	While detecting LP_IDLE on XLGMII or CGMII	LPI:M	Yes [] N/A []

81.5.3.7 Link Interruption

Item	Feature	Subclause	Value/Comment	Status	Support
*LINT1	Detection of Link Interruption	81.3.4	As defined in Table 81–5	O	Yes [] No []
LINT2	CARRIER_STATUS response to Link Interruption	81.1.7.3	Set to CARRIER_ON if link_fault is Link Interruption	LINT1: M	Yes [] N/A []